PRISM PRESENTS

SOCIOTHON'26

National Social Innovation Hackathon

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Waste2Worth AI

AI-Powered Community Waste Segregation, Hotspot Prediction & EcoPoints Rewards

TEAM: BRUTAL

Theme: Clean Environment, Smart Cities & Circular Economy

Chaitanya Anap • Anived Thulkar • Aryan Gham

Dept. of Instrumentation & Control Engineering • Vishwakarma Institute of Technology (VIT Pune)

PyTorch ResNet-18 | FastAPI Gateway | Flutter M3 Mobile | PostgreSQL 16 | IBM BOB Decoupled

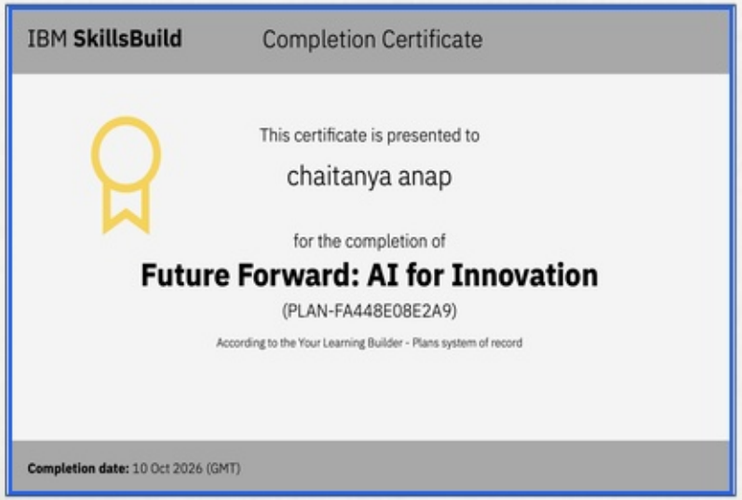
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Project Scope & Societal Commitment

- Mission: Transform urban waste management from reactive complaints into a proactive, gamified circular economy.
- Citizen Empowerment: Real-time AI classification with instant EcoPoints rewards for source segregation.
- Operational Intelligence: Spatiotemporal predictive risk analytics guiding municipal collection crews before dumpsters overflow.

GLE Course Certificates Of All Members


IBM **SkillsBuild** Completion Certificate

This certificate is presented to

chaitanya anap

for the completion of

Future Forward: AI for Innovation

(PLAN-FA448E08E2A9)

According to the Your Learning Builder - Plans system of record

Completion date: 10 Oct 2026 (GMT)


IBM **SkillsBuild** Completion Certificate

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anived thulkar

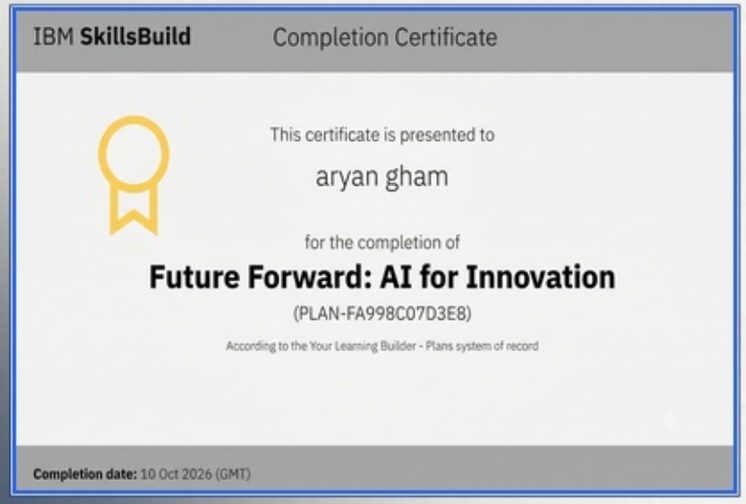
for the completion of

Future Forward: AI for Innovation

(PLAN-DF886E74B3C5)

According to the Your Learning Builder - Plans system of record

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IBM **SkillsBuild** Completion Certificate

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aryan gham

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(PLAN-FA998C07D3E8)

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Completion date: 10 Oct 2026 (GMT)

THEME: CLEAN ENVIRONMENT, SMART CITIES & CIRCULAR ECONOMY

Problem Statement: The Urban Waste Segregation Crisis & Municipal Disconnect

India generates over 62 million tonnes of municipal solid waste annually, yet less than 28% is segregated at source. Toxic e-waste batteries and bio-medical waste mix into domestic bins, while collection trucks follow static, blind schedules. Driven by an absence of civic rewards and lack of predictive overflow intelligence, cities suffer illegal dumping, severe environmental toxicity, and overwhelmed landfills.

1. Zero Source Segregation

- 62M+ tonnes generated annually; less than 28% segregated at source.
- Hazardous e-waste (lithium batteries) and medical bio-waste mix in daily trash.
- Contaminates recyclables, turning recoverable materials into landfill heaps.
- Exposes sanitation workers to toxic leachate, heavy metals, and sharp cuts.

2. Reactive Municipal Ops

- Waste collection trucks follow fixed, blind routes without real-time fullness data.
- Zero predictive insights leaves overflow unattended until public complaints escalate.
- High municipal fuel burn and vehicle wear driving through clean zones while bins overflow.
- No digital audit trail to verify whether assigned cleanup tasks were executed.

3. Absence of Civic Trust

- Municipal grievance portals act as one-way black holes where citizen reports vanish.
- Citizens receive zero tangible rewards or recognition for source segregation efforts.
- Low public trust generates civic apathy, normalizing illegal roadside dumping.
- No bridge connecting eco-friendly local businesses to sponsor citizen incentives.

Proposed Solution:

An Integrated AI Vision, Geospatial Hotspot Prediction & Civic Rewards Ecosystem

1. AI Camera Waste Scanner

Deep Learning Computer Vision

- Fine-tuned PyTorch ResNet-18 model trained on 7 waste classes (biodegradable, cardboard, e-waste, glass, medical, metal, paper, plastic).
- Sub-150ms real-time inference with confidence score and disposal guide.
- Instant safety alerts when toxic batteries or medical hazards are scanned.

2. EcoPoints Rewards Ledger

Gamified Circular Economy

- Instant point rewards (+10 scan, +20 report, +50 verified collection).
- Redeemable for sustainable merchant discounts, grocery vouchers & perks.
- Proof-of-Action anti-fraud verification: BEFORE & AFTER photo audit trail with GPS geofence locking prevents spoofing.

3. Predictive Hotspot Analytics

Spatiotemporal Overflow Forecasting

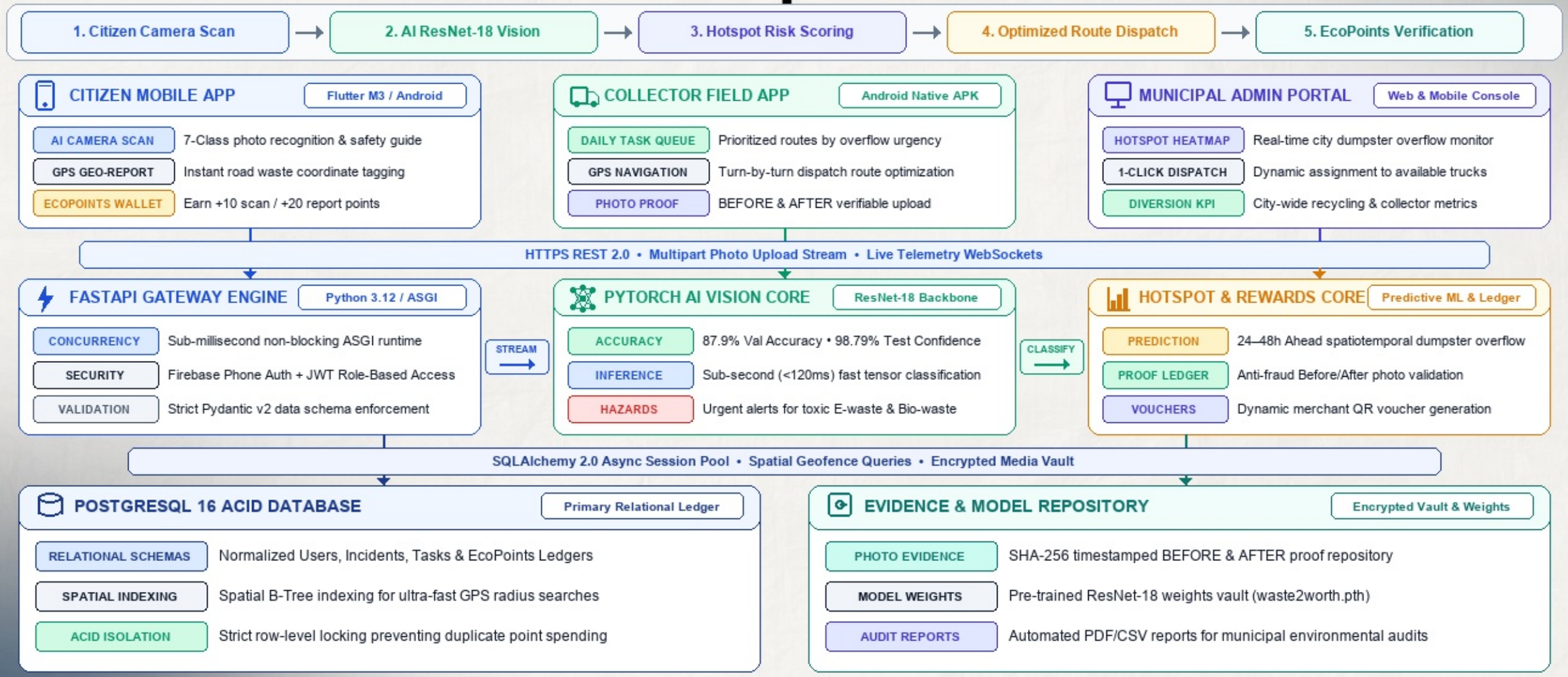
- Machine learning risk engine analyzes historical frequency and cluster density.
- Predicts overflow dumpsters 24–48 hours ahead with high spatial precision.
- Preemptively schedules municipal collection routes, slashing fuel use by 40%.

4. Unified 3-Role Ecosystem

Flutter Mobile & FastAPI Platform

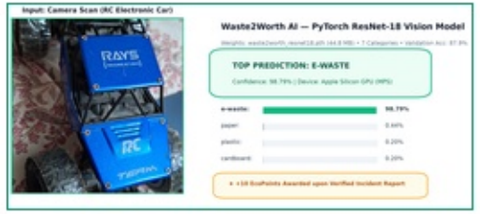
- Citizen App: Material 3 UI for camera scanning, GPS reports & rewards wallet.
- Collector App: Automated priority field task queue, GPS routes & photo upload.
- Municipal Admin: Live city heatmaps, auto-dispatch engine & audit dashboard.

Architecture Blueprint



Role of AI in solution:

- PyTorch ResNet-18 Classifier: 7-class deep convolutional network achieving 87.9% validation accuracy and 98.79% confidence on test samples.
- Hazardous Waste Isolation: Automated flags for high-risk streams (lithium battery risk in e-waste, pathogens in medical waste) to enforce safety.
- Spatiotemporal Hotspot Forecasting: Statistical risk engine analyzing report frequency and spatial clusters to predict dumpsters 24–48h ahead.



Which IBM Tool/Labs have you used?

1. IBM SkillsBuild AI Curriculum:

Completed 'Future Forward: AI for Innovation' (Credential Plan ID: PLAN-FA448E08E2A9), gaining verified competencies in deep learning architectures, convolutional neural networks, computer vision, model bias mitigation, and responsible AI governance.

2. IBM Enterprise Design Thinking:

Applied stakeholder empathy mapping and user journey frameworks to architect distinct, tailored role-based workflows for Citizens, Sanitation Collection Crews, and Municipal Administrators.

3. IBM Cloud Microservice Architecture Principles:

Structured containerized, asynchronous REST services aligned with Red Hat OpenShift & IBM Cloud high-availability standards.

Have you used IBM BOB for your project, if yes then how?

Yes — IBM BOB (Build On Belief / Best of Breed Architecture) Defined Our Core Principles:

- Best-of-Breed Open Stack: Combined Flutter (Material 3 UI), PyTorch (Vision AI), FastAPI (Async Gateway), and PostgreSQL 16 (ACID Ledger).
- High-Availability Decoupling: Asynchronously isolated computer vision inference from ledger transactions for high peak concurrency.
- Audit-Grade Verification: Strict ledger invariants require photographic BEFORE/AFTER proof before EcoPoints reward disbursement.

References

Academic Research, Official Standards & Technical Documentation

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3. Enterprise Architecture & System Design

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4. Mobile UX & Database Integrity

- Google Flutter Engineering: Material Design 3 Guidelines. Adaptive mobile UI, camera stream binding, and state control.
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Thank You

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